

Parking-Lot Stormwater Detention

From project idea to simulation,
design selection, and engineering evaluation

Prepared for Manasi

Project: manasi_project

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Abstract

This project investigates how temporary storage can reduce the peak discharge from a hypothetical parking lot while meeting explicit overflow and drainage targets. The study models a 4,000 m² parking catchment and an adjacent 200 m² detention footprint. Nine synthetic one-hour storms combine rainfall depths of 25, 50, and 100 mm with uniform, front-loaded, and back-loaded distributions. Event runoff is calculated using cumulative curve-number losses, transformed into an inflow hydrograph using a volume-normalized SCS response, and routed through constant-area storage with an idealized square outlet. An implicit numerical solution conserves water and explicitly records overflow.

The design search evaluates 24 storage–outlet combinations against all nine storms. The smallest tested design satisfying all three 50 mm storm shapes provides 150 m³ of storage at 0.75 m modeled depth with a 100 × 100 mm opening. Its peak reduction ranges from 63.09% to 74.39%, with zero overflow and a maximum drawdown time of 7.81 hours after rainfall ends. Verification includes 16 passing numerical tests, water-balance checks, and refinement of the complete grid to a 1.875 s time step.

The selected design overflows during the 100 mm storms and fails one CN 100 sensitivity case. The findings therefore support a conditional conceptual selection under the stated assumptions. The modeled benefit is reduced peak flow and delayed discharge; detention does not remove the incoming runoff volume.

Selected nominal configuration 150 m³ storage · 200 m² footprint · 0.75 m depth
100 × 100 mm opening · minimum design-event peak reduction 63.09%

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Reading guide. Sections 1–2 establish the project idea and design objectives. Sections 3 and 4 explain how the study was performed. Sections 5–7 present the measured simulation outputs and their numerical checks. Section 8 interprets the design decision and its limits. The appendices provide the complete candidate comparison and instructions for reproducing the work.

Evidence convention. Numerical results in this report come from the saved project outputs, with copies in `source_data/`. Tables are generated directly from those copies. The cited engineering manuals support the method; the geometry, synthetic rainfall, parameter choices, and acceptance targets are project assumptions. Values are rounded for presentation, while design decisions use the saved full-precision values.

1 Project Idea and Engineering Goals

1.1 Origin of the project

The project began with a practical engineering question: can a compact detention facility temporarily store parking-lot runoff and release it slowly enough to reduce the downstream peak? The supplied project plan framed this as a manageable study connecting fluid mechanics, numerical methods, Python analysis, spreadsheet checks, and conceptual drafting.^[1] Its purpose was to develop an engineering argument from stated assumptions through calculations and comparison to a defensible design choice.

A parking lot provides a simple catchment with a high assumed runoff response. During a storm, runoff can arrive faster than a restricted outlet can release it. A detention basin accommodates the difference as stored water. Water depth then controls discharge through the opening. The resulting tradeoff is central to the project: a smaller opening can suppress controlled flow but may increase storage requirements, overflow, and drainage time. Capacity and outlet size must therefore be evaluated together.

The study uses a hypothetical site so the modeling assumptions can be examined explicitly and reproduced without field data. Its results describe the behavior of that conceptual system. No measured flooding history, surveyed drainage pattern, or location-specific storm frequency was supplied.

1.2 Primary question and acceptance criteria

The primary question is:

What is the smallest tested storage capacity that reduces the peak downstream discharge by at least 50%, avoids overflow, and drains below 1% of capacity within 24 hours after the selected storm ends?

The design event is 50 mm of rain in 60 minutes. A candidate must meet every criterion for *each* of the three temporal storm shapes. The criteria are educational targets fixed before comparison, rather than adopted local drainage requirements.

Table 1: Project objectives and the quantities used to assess them.

Objective	Assessment rule
Peak attenuation	At least 50% reduction relative to the matching baseline; downstream flow includes controlled discharge and overflow.
Overflow avoidance	Total overflow no greater than 10^{-6} m^3 , the specified numerical tolerance.
Drainage recovery	Final crossing below 1% of nominal capacity within 24 h after rain ends, with no later rebound above the threshold.
Storage economy	Choose the lowest-capacity passing option in the tested grid.
Reliable computation	Check water balance, limiting cases, and time-step refinement before interpreting the result.

Secondary goals are to explain why alternatives fail, examine performance at 25 and 100 mm rainfall, test selected assumptions, and produce a reproducible set of calculations, figures, and engineering deliverables. Cost, land value, and construction quantities are not included in the optimization objective.

2 Site Concept and Design Basis

2.1 Geometry and a consistent baseline

The parking catchment measures 80×50 m, giving a contributing area $A_p = 4,000$ m². The adjacent basin footprint measures 20×10 m, giving a fixed area $A_b = 200$ m². The total study area is therefore 4,200 m² for both the baseline and detention alternatives.

In the baseline, the reserved basin footprint is represented as an impervious collection pad with negligible storage and loss. Its rainfall passes directly to the downstream comparison point. In a detention alternative, rainfall on that same footprint enters storage together with parking runoff. This keeps the contributing area and prescribed event input volume consistent. Omitting the footprint from the baseline would change the comparison itself.

Table 2: Nominal inputs and the design search space.

Quantity	Adopted value
Parking area; basin footprint	4,000 m ² ; 200 m ²
Curve number; initial abstraction	CN = 98; $I_a = 0.2S$
Catchment lag	10 min
Storm depths and duration	25, 50, 100 mm; 60 min for each event
Storm distributions	Uniform, front-loaded, back-loaded; six 10-min blocks
Candidate depths	0.15, 0.30, 0.45, 0.60, 0.75, 0.90 m
Candidate storage capacities	30, 60, 90, 120, 150, 180 m ³
Square opening side lengths	40, 60, 80, 100 mm
Outlet coefficient and gravity	$C_d = 0.62$; $g = 9.80665$ m/s ²
Initial basin storage	0 m ³ at the start of each independent event
Basin infiltration; evaporation	Both zero
Time step	Initially 30 s; final accepted step 1.875 s
Simulation horizon	Initially 30 h; extend by doubling if required, up to the configured 240 h cap

Source: Bundled `source_data/config_used.json`.

The basin follows a constant-area stage–storage relationship, $V = A_b h$. Thus, an increase in modeled depth increases capacity without changing the footprint. Vertical boundaries in the drawing illustrate this accounting geometry; the study does not size structural walls or prescribe an excavated basin with safe side slopes.

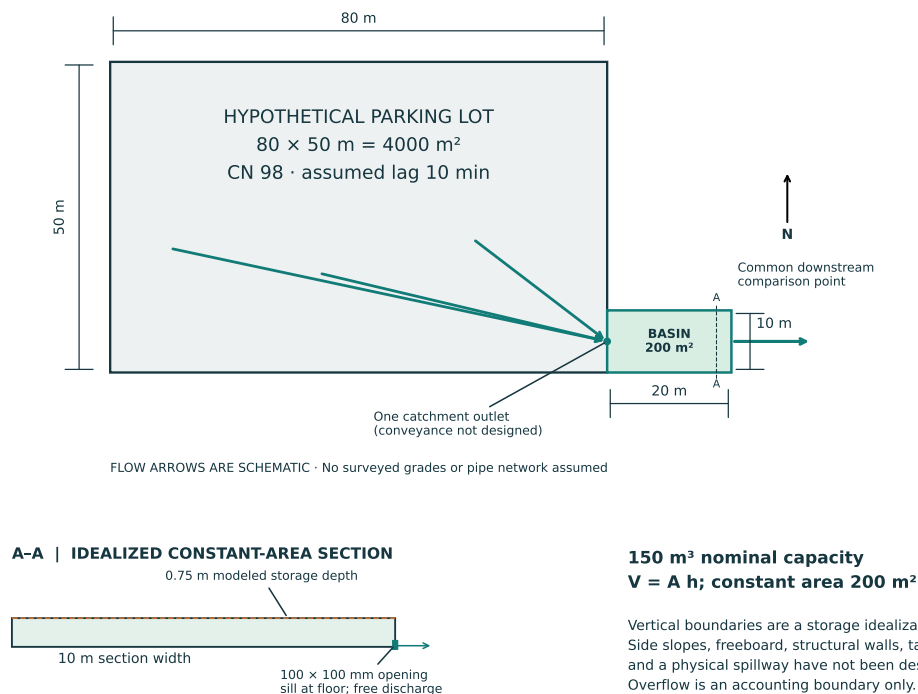
2.2 Scope of the engineering representation

The system contains one lumped parking catchment, one inflow hydrograph, one level-pool storage element, and one freely discharging control opening. The adopted CN, lag, and discharge coefficient are preliminary assumptions. The storms have no assigned recurrence intervals. Tailwater, pipe friction, debris blockage during the design events, water-quality treatment, groundwater, and sequences of storms with incomplete recovery are not modeled.

These choices isolate the interaction between rainfall timing, runoff, storage, and outlet discharge. They also define the practical boundary of the finding: a passing simulated configuration is a conceptual candidate for further development, rather than a completed site drainage design.

PARKING-LOT DETENTION / CONCEPTUAL LAYOUT

Same 4200 m² contributing study area in all cases · Dimensions in metres · No construction design



DRAWING 01 · Diagrammatic plan and section; use labeled dimensions. CAD units: 1 drawing unit = 1 metre.

Figure 1: Conceptual site plan and basin section from the completed project. The labeled dimensions define the hypothetical geometry. Flow arrows are schematic; surveyed grades, conveyance, freeboard, and a physical spillway are outside the model.

3 Engineering Methods

3.1 Rainfall construction

Each one-hour event contains six equal ten-minute blocks. For total rainfall P_{tot} and relative block weights r_k , block depth is

$$P_k = P_{\text{tot}} \frac{r_k}{\sum_{j=1}^6 r_j}. \quad (1)$$

The vectors are (1, 1, 1, 1, 1, 1) for uniform rainfall, (6, 5, 4, 3, 2, 1) for front-loaded rainfall, and (1, 2, 3, 4, 5, 6) for back-loaded rainfall. For the 50 mm event, uniform intensity is 50 mm/h throughout the hour. The front-loaded intensities are approximately 85.71, 71.43, 57.14, 42.86, 28.57, and 14.29 mm/h; the back-loaded sequence reverses that order.

The implementation integrates the constant-intensity blocks over each computational interval, including intervals that cross a block boundary. It verifies that rainfall increments sum to the specified event depth. Storm shapes change the timing of rainfall while preserving total depth and duration.

3.2 Cumulative runoff losses

The NRCS/SCS curve-number method calculates cumulative excess rainfall from cumulative precipitation. With P , S , I_a , and P_e in millimetres,

$$S = \frac{25400}{\text{CN}} - 254, \quad I_a = 0.2S, \quad (2)$$

$$P_e(P) = \begin{cases} 0, & P \leq I_a, \\ \frac{(P - I_a)^2}{P - I_a + S}, & P > I_a. \end{cases} \quad (3)$$

These equations and the event-based interpretation follow the USACE description of the SCS CN loss model.[2] For an interval bounded by t_n and t_{n+1} , the runoff depth is

$$\Delta P_{e,n} = P_e(P(t_{n+1})) - P_e(P(t_n)), \quad v_{p,n} = \frac{A_p \Delta P_{e,n}}{1000}. \quad (4)$$

Here $v_{p,n}$ is generated parking-runoff volume in m^3 . Initial abstraction is applied once through the cumulative event calculation; it is not restarted at every interval. The event state resets between independent storms. CN 100 is handled explicitly as $P_e = P$, including at zero rainfall. The parking area receives no additional impervious-area multiplier.

For the nominal 50 mm event, the worked calculation is

$$\begin{aligned} S &= 5.183673 \text{ mm}, & I_a &= 1.036735 \text{ mm}, \\ P_e &= 44.275843 \text{ mm}, & V_{\text{parking}} &= 177.103371 \text{ m}^3. \end{aligned}$$

Direct rainfall on the basin footprint adds $200(50/1000) = 10.000000 \text{ m}^3$, so total input to either alternative is 187.103371 m^3 . This scalar check supplies an independent reference for the time-series water balance.

3.3 Runoff hydrograph transformation

Detention routing needs the time history of inflow as well as its total volume. A peak-only estimate cannot resolve temporary storage through the event. TxDOT's runoff-method guidance identifies hydrograph methods for storage applications.[7]

The project uses the 33 dimensionless SCS/NRCS hydrograph ordinates in TxDOT Table 4-29, stored locally as `scs_dimensionless_uh.csv`.^[4] For each excess-rainfall interval, the response time to peak is

$$T_p = t_{\text{lag}} + \frac{\Delta t}{2}, \quad (5)$$

measured from the beginning of the pulse. This follows the SCS unit-hydrograph timing convention.^[3]

The implementation linearly interpolates the dimensionless response $f(t/T_p)$ and integrates it over each output interval. It converts these integrals to nonnegative volume fractions w_j satisfying

$$w_j = \frac{\int_{j\Delta t}^{(j+1)\Delta t} f(t/T_p) dt}{\int_0^{5T_p} f(t/T_p) dt}, \quad \sum_j w_j = 1, \quad (6)$$

where f is zero beyond the reference tail. The transformed parking inflow is the discrete convolution

$$I_{p,n} = \sum_j v_{p,n-j} w_j. \quad (7)$$

The complete response through $5T_p$ is retained. Normalization makes the integrated parking hydrograph equal the generated parking-runoff volume. The tabulated dimensionless discharge values are used as a response shape, rather than interpreted directly as dimensional flow.

Direct footprint rainfall enters without this catchment-lag transformation. For rainfall increment ΔP_n , interval input and baseline flow are

$$I_n = I_{p,n} + \frac{A_b \Delta P_n}{1000}, \quad Q_{b,n} = \frac{I_n}{\Delta t}. \quad (8)$$

Here I_n is a volume in m^3 , while $Q_{b,n}$ is an interval-average baseline flow in m^3/s .

3.4 Storage and outlet hydraulics

For fixed basin area, water depth is $h = V/A_b$. The opening is a vertical square with side a , with its sill at the basin floor. Under the project's free-discharge idealization, local velocity depends on the water head above each wetted strip of the opening. Integrating across the wetted height gives

$$\begin{aligned} Q_c(h) &= \int_0^{\min(h,a)} C_d a \sqrt{2g(h-z)} dz \\ &= \frac{2}{3} C_d a \sqrt{2g} \left[h^{3/2} - \max(h-a, 0)^{3/2} \right]. \end{aligned} \quad (9)$$

All lengths in this equation are in metres and discharge is in m^3/s . The expression approaches zero continuously at an empty basin and accounts for an opening that is only partly wetted at shallow depth.

For water well above the opening, the conventional centroid-head estimate $Q_c \approx C_d a^2 \sqrt{2g(h-a/2)}$ provides a useful independent comparison. USACE documentation distinguishes outlet flow regimes and the limits of submerged-orifice equations.^[6] Equation (9) is the project's assumed rating, with constant $C_d = 0.62$; it is not a field-calibrated outlet or an assertion of equivalence to a particular HEC-HMS structure model.

3.5 Implicit storage routing and overflow accounting

The governing conservation equation is

$$\frac{dV}{dt} = Q_{\text{in}} - Q_c - Q_o, \quad (10)$$

where Q_o is overflow discharge. The level-pool conservation concept follows reservoir-routing practice.[5] The code uses a backward-Euler storage step, whereas the cited HEC-HMS discussion describes a trapezoidal Modified Puls formulation.

For a nonoverflowing interval, the next storage value satisfies

$$V_{n+1} + \Delta t Q_c(V_{n+1}/A_b) = V_n + I_n. \quad (11)$$

The code solves this scalar equation with SciPy's bracketed Brent solver on $[0, \min(V_{\text{max}}, V_n + I_n)]$. The left side increases with storage, so the bracket contains a unique physical solution when capacity is sufficient. The controlled discharge assigned to the interval is the endpoint rating $Q_c(V_{n+1}/A_b)$, whose product with Δt supplies the step's numerical discharge volume.

Let $W_n = V_n + I_n$ and $Q_{\text{cap}} = Q_c(V_{\text{max}}/A_b)$. If $W_n \geq V_{\text{max}} + \Delta t Q_{\text{cap}}$, the update becomes

$$V_{n+1} = V_{\text{max}}, \quad Q_{c,n} = Q_{\text{cap}}, \quad Q_{o,n} = \max\left(\frac{W_n - V_{\text{max}}}{\Delta t} - Q_{\text{cap}}, 0\right). \quad (12)$$

Excess water is consequently retained in the accounting as overflow to the same downstream point. It is not discarded by clipping the storage series. The downstream discharge used in every comparison is $Q_{d,n} = Q_{c,n} + Q_{o,n}$. Overflow is a capacity boundary calculation; no spillway geometry or inundation hydraulics is represented.

3.6 Performance metrics and numerical conventions

Peak attenuation is calculated as

$$R_Q = 100 \left(1 - \frac{\max_n Q_{d,n}}{\max_n Q_{b,n}} \right). \quad (13)$$

Each comparison uses the baseline generated with the same storm and catchment parameters. Maximum depth follows from maximum storage divided by A_b . The model also records overflow volume, discharge volume, residual storage, time to peak, and drainage time.

Flows use interval values with midpoint timestamps; storage uses interval boundaries. Therefore, flow volume is obtained by summing flow times Δt , rather than trapezoidal integration of midpoint samples. Overflow onset is recorded at the start of its first nonzero interval.

Drawdown is the elapsed time from rainfall end to the final interpolated crossing below $0.01V_{\text{max}}$, after which storage remains below that threshold. For the selected capacity, this is 1.50 m^3 , equivalent to 7.50 mm depth. Exact emptying is not required: at shallow depth, Equation (9) has $Q_c \propto h^{3/2}$, producing an asymptotic recession. Any storage still present at the reporting horizon is included in the water balance.

4 Implementation and Experimental Workflow

4.1 Computational organization

The project is implemented as a local Python workflow. Inputs are separate from the hydrology, routing, comparison, and reporting modules, allowing an assumption to be changed without rewriting the physical model.

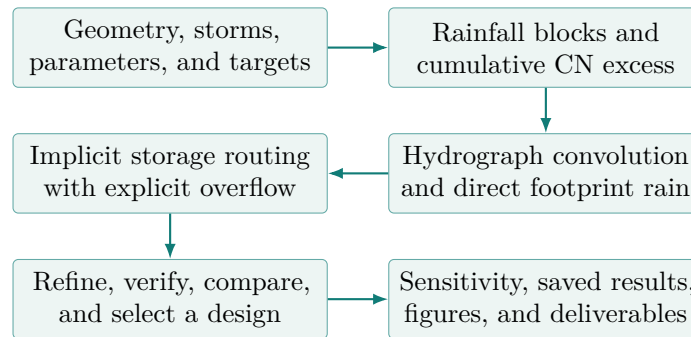


Figure 2: Implemented engineering workflow. All numerical storm inputs and reference hydrograph ordinates are available locally.

Table 3: Main source modules and their responsibilities.

Source file	Responsibility
<code>run_project.py</code>	Validate inputs, run tests and refinement, orchestrate analysis and deliverable generation.
<code>stormwater/hydrology.py</code>	Rainfall increments, cumulative CN runoff, unit-hydrograph weights, and baseline events.
<code>stormwater/routing.py</code>	Outlet rating, implicit storage solution, overflow, and final drainage crossing.
<code>stormwater/study.py</code>	Scenario metrics, design selection, convergence comparison, sensitivity, and independent checks.
<code>stormwater/figures.py</code>	Scientific plots and conceptual site drawings.
<code>stormwater/workbook.py</code>	Workbook tables, independent formulas, checks, and charts.
<code>stormwater/dashboard.py</code>	Offline interactive scenario exploration.
<code>tests/test_model.py</code>	Hydrologic and hydraulic numerical tests.

4.2 Design sweep and fixed selection rule

Six capacities and four square opening sizes give 24 candidate designs. Three rainfall depths and three storm shapes give nine events. Each design is run against each event, giving $24 \times 9 = 216$ routed cases, plus nine matching baselines at each time resolution.

Selection considers only the three 50 mm events. A design is feasible when all three shapes satisfy peak reduction, overflow, and drawdown requirements. Feasible designs are ranked first by increasing storage capacity, then by shortest worst-case drawdown, least worst-case overflow, and finally largest opening. This fixed order makes the decision reproducible and avoids selecting an attractive hydrograph while overlooking a failing constraint.

The complete grid is repeated at successively halved time steps. The run requires at least three resolutions, checks relative changes in downstream peak, baseline peak, and maximum storage, and continues refinement until the 1% target and decision-stability checks are met. The initial 30 h horizon is extended when drainage has not been established; the longest horizon needed in the completed core study was 60 h.

4.3 Sensitivity experiment

After selection, the chosen capacity and opening are held fixed. The model varies CN through 95, 98, and 100; lag through 5, 10, and 15 min; and C_d through 0.50, 0.62, and 0.75. Each value is run for all three 50 mm storm shapes, giving 27 rows, including nominal repeats. Each comparison recomputes its matching baseline. These are one-at-a-time experiments, rather than joint worst-case combinations or measured parameter uncertainty distributions.

5 Results

5.1 Runoff volume and baseline response

At a fixed CN, storm shapes with the same total rainfall have the same calculated runoff volume. Their flow peaks differ because the temporal distribution and hydrograph response differ.

Table 4: Event water inputs at nominal CN 98, common to every storm shape and detention capacity for a given rainfall depth.

Rainfall (mm)	Parking excess (mm)	Parking runoff (m ³)	Direct rain (m ³)	Total input (m ³)
25	19.701	78.806	5.000	83.806
50	44.276	177.103	10.000	187.103
100	94.038	376.150	20.000	396.150

Source: Bundled `source_data/baselines.csv`.

For the design event, the baseline peak is 57.48 L/s for uniform rainfall, 73.19 L/s for front-loaded rainfall, and 85.67 L/s for back-loaded rainfall. The highest baseline peak therefore occurs in the back-loaded case even though all three events deliver the same 187.103 m³ to the system.

5.2 Selected capacity and outlet

The smallest tested feasible design is **150 m³ of nominal storage at 0.75 m depth over 200 m², with a 100 × 100 mm opening**. Table 5 reports its performance for the three design storm shapes.

Table 5: Selected-design performance for the 50 mm storms. Drawdown is measured after rainfall ends.

Shape	Baseline (L/s)	Detained (L/s)	Reduction (%)	Max. storage (m ³)	Overflow (m ³)	Drawdown (h)
Uniform	57.48	21.22	63.09	129.48	0.000	7.65
Front-loaded	73.19	21.04	71.25	127.55	0.000	7.51
Back-loaded	85.67	21.94	74.39	137.76	0.000	7.81

Source: Bundled `source_data/scenarios.csv`, design h0.750_o100, final time step 1.875 s.

The controlling storm depends on the metric. Uniform rainfall gives the *lowest percentage peak reduction*, 63.09%, because its baseline peak is lower. Back-loaded rainfall gives the *greatest absolute detained peak and storage demand*, 21.94 L/s and 137.761 m³, together with the longest drawdown, 7.806 h. Thus, the most demanding storage event need not be the event with the smallest percentage attenuation.

The maximum design-event water depth is 0.6888 m. The difference between nominal capacity and maximum storage is 12.239 m³, or approximately 0.0612 m of depth over the fixed footprint. This is unused modeled capacity for these cases; it is not an engineered freeboard allowance.

5.3 Why other candidates fail

Four of the 24 tested designs pass all three nominal design storms. Every capacity below 150 m³ fails. At 120 m³, even the 100 mm opening, which has the least worst-case overflow among that capacity's tested openings, produces 18.459 m³ of overflow and a worst peak reduction of only 7.49%.

At the selected 150 m³ capacity, the 80 mm opening achieves at least 66.92% peak reduction but produces 1.441 m³ of worst-case overflow, so only two of the three storm shapes pass. The 100 mm opening provides enough release during the event to avoid overflow while retaining adequate peak attenuation.

At 180 m³, the 60, 80, and 100 mm openings pass. The 40 mm opening achieves 92.96% worst-case peak attenuation without overflow but requires 28.84 h for drawdown, exceeding the 24 h limit. These cases demonstrate why a single peak-reduction objective would produce an incomplete decision.

5.4 Hydrograph attenuation and timing

Figure 4 shows how the basin changes the response. Parking runoff and direct rainfall create a relatively short inflow event; the basin stores part of that input and releases it over a longer period. The controlled-outlet peak is approximately 21–22 L/s for all three design shapes, despite the wider range in baseline peaks.

Table 6: Time to peak from rainfall start for the selected design events.

Shape	Baseline peak (min)	Detained peak (min)	Peak delay (min)
Uniform	59.98	73.39	13.41
Front-loaded	26.23	64.58	38.34
Back-loaded	59.98	76.17	16.19

Source: Bundled `source_data/scenarios.csv`; interval-midpoint times.

The detained peak occurs after rainfall ends in every design case. The front-loaded case has the largest shift between baseline and detained peaks, about 38.34 min. These timing changes describe the modeled

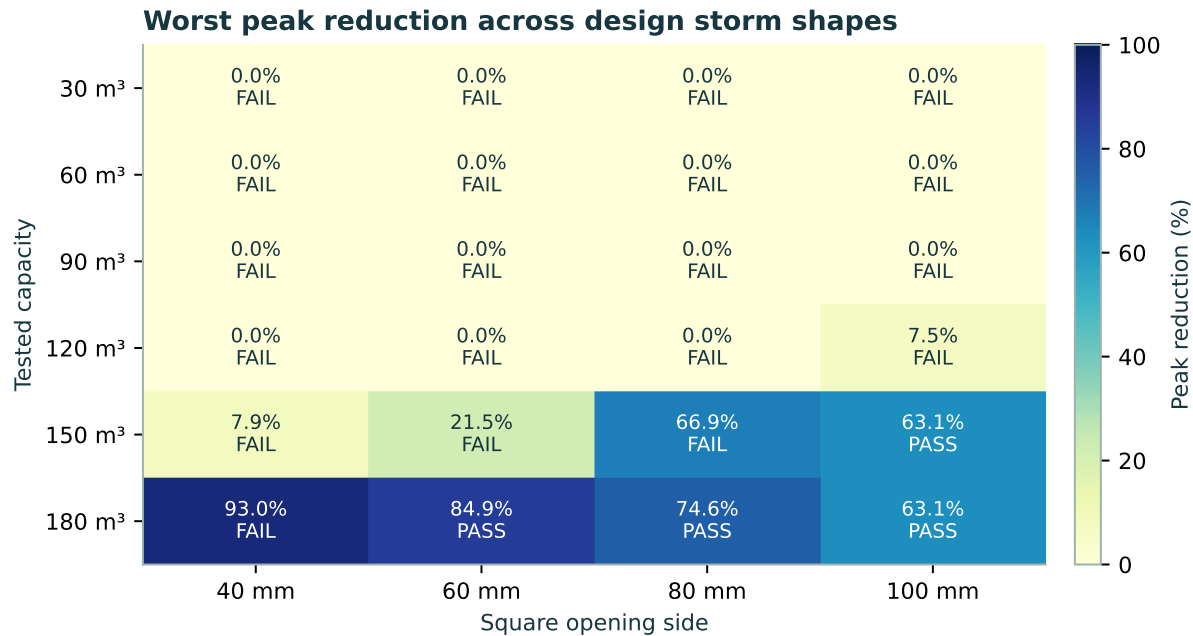


Figure 3: Comparison of all 24 designs for the 50 mm storms. Cell values show the smallest peak reduction across the three shapes. PASS requires all three criteria for every shape. The full numerical table is provided in Appendix A.

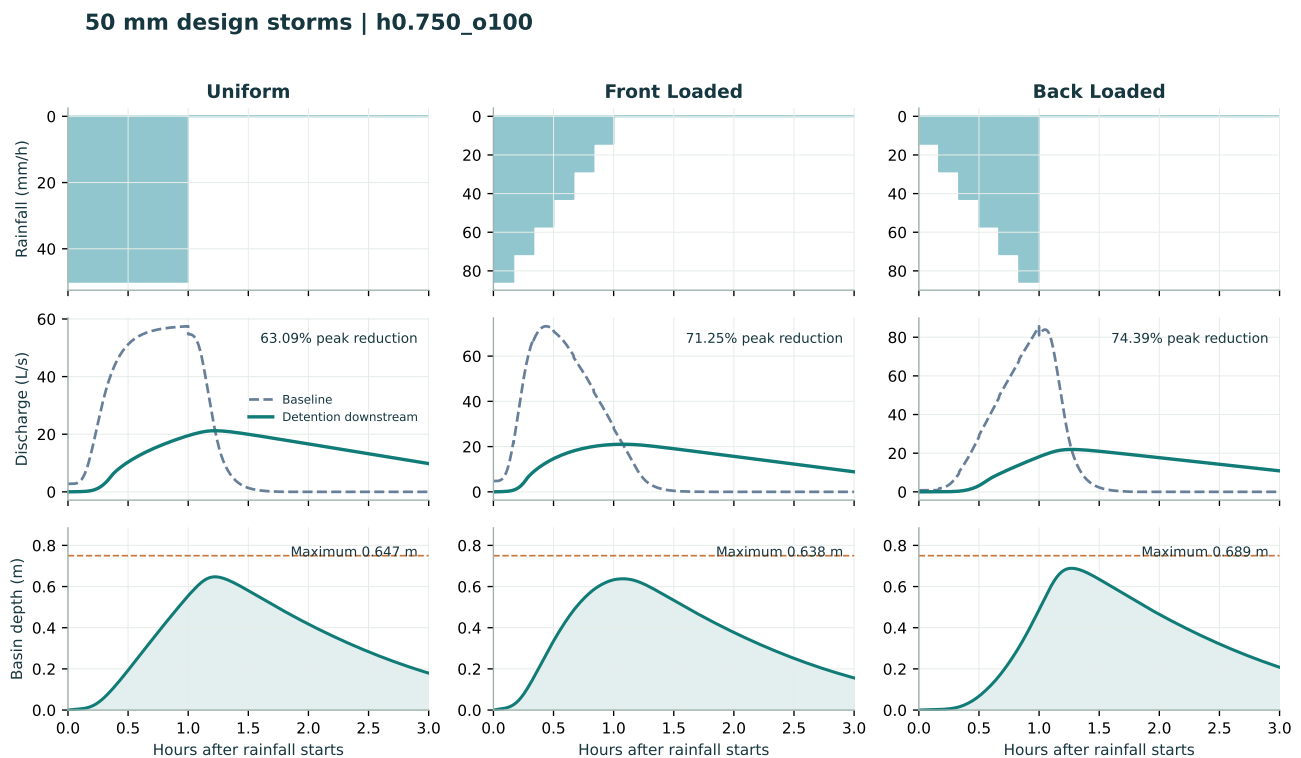


Figure 4: Rainfall, baseline and detained discharge, and basin depth for the selected design under the 50 mm storms. The dashed depth line marks nominal capacity. Curves show the first three hours; the drainage tail continues beyond this window.

comparison point; the study does not route the discharge through a downstream channel network.

5.5 Performance outside the design event

The selected geometry is unchanged for the 25 and 100 mm tests. Table 7 presents all nine events together.

Table 7: Selected-design results across all tested rainfall depths.

Rain (mm)	Shape	Baseline (L/s)	Detained (L/s)	Reduction (%)	Overflow (m ³)	Drawdown (h)
25	Uniform	27.77	12.77	54.02	0.000	6.39
25	Front-loaded	32.68	12.52	61.69	0.000	6.23
25	Back-loaded	40.90	13.45	67.12	0.000	6.53
50	Uniform	57.48	21.22	63.09	0.000	7.65
50	Front-loaded	73.19	21.04	71.25	0.000	7.51
50	Back-loaded	85.67	21.94	74.39	0.000	7.81
100	Uniform	116.19	116.19	0.00	162.526	7.97
100	Front-loaded	154.83	149.86	3.21	163.580	7.87
100	Back-loaded	173.96	173.96	0.00	173.330	8.02

Source: Bundled `source_data/scenarios.csv`. Numerically negligible positive or negative reductions are displayed as 0.00%.

The 25 mm events produce no overflow and show peak reductions of 54.02–67.12%. The 100 mm events fill the basin and generate 162.526–173.330 m³ of overflow. Their downstream peak reductions are approximately 0–3.21%, because overflow carries much of the inflow past the storage limit. Reporting the controlled outlet alone would conceal this failure. The larger events are stress tests with no assigned return period.

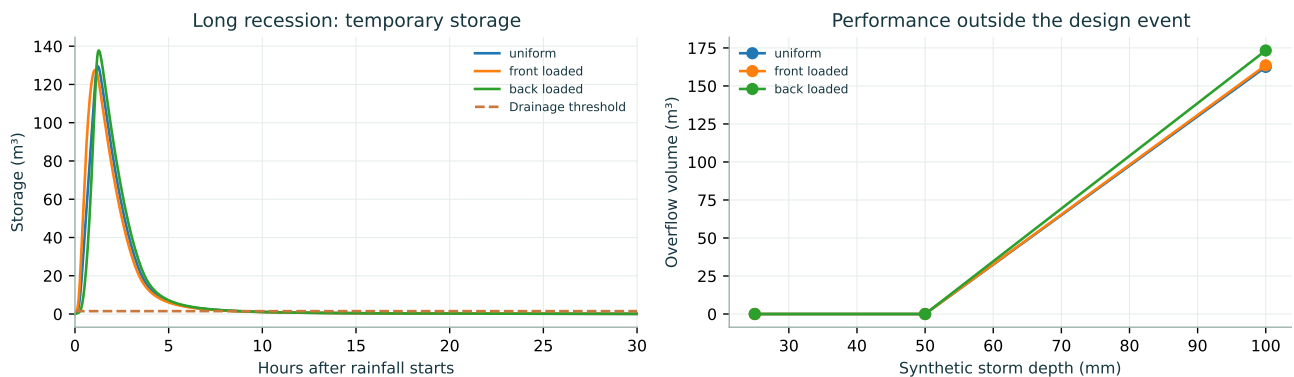


Figure 5: Selected-design recession for the 50 mm storms and overflow volume across rainfall depths. The drainage threshold is 1.50 m³. The overflow panel connects the three tested depths for visual comparison; intermediate storm depths were not simulated in this study.

5.6 Water budget and the meaning of detention

With zero initial storage, basin infiltration, and evaporation, conservation requires

$$V_{\text{input}} = V_{\text{controlled}} + V_{\text{overflow}} + V_{\text{residual}} + \varepsilon_V, \quad (14)$$

where ε_V is the numerical balance residual. The selected 50 mm events use a 30 h horizon. Their small remaining volumes are shown explicitly.

Table 8: Selected-design water budget at 30 h after rainfall starts. “Discharged” here denotes controlled outlet volume; overflow is separate.

50 mm shape	Input (m ³)	Discharged (m ³)	Overflow (m ³)	Residual (m ³)
Uniform	187.103371	187.011722	0.000000	0.091649
Front-loaded	187.103371	187.012613	0.000000	0.090759
Back-loaded	187.103371	187.010723	0.000000	0.092649

Source: Bundled `source_data/scenarios.csv`. Display rounding can produce a final-digit difference in the sum.

Almost the entire inflow volume has been released by the horizon, with about 0.091–0.093 m³ remaining. Detention changes when water is discharged; it does not eliminate the 187.103 m³ entering storage. Catchment losses in the CN calculation occur before the basin and are the same in both alternatives. They are not an additional volume-reduction benefit of the basin.

6 Verification and Numerical Reliability

6.1 Boundary tests and independent calculations

The numerical test suite contains 16 tests, all of which passed in the recorded project run and passed again during preparation of this report. They cover rainfall totals and partial blocks, zero and sub-abstraction rainfall, CN 100, independent CN arithmetic, hydrograph volume and timing, invalid inputs, empty storage, blocked outlets, insufficient capacity, outlet quadrature, deep-orifice comparison, monotonicity, shallow drainage, final drainage crossing, and stepwise continuity.

For the outlet rating, independent numerical quadrature agrees with the integrated formula to a maximum absolute difference of approximately 2.08×10^{-17} m³/s in the saved checks. This tests the implementation of the idealized equation; it does not calibrate the physical coefficient.

When the basin is shallow enough that $h < a$ and no inflow remains, define $k = (2/3)C_d a \sqrt{2g}$. Then the model reduces to $A_b dh/dt = -kh^{3/2}$, with analytical solution

$$h(t) = \left[h_0^{-1/2} + \frac{k}{2A_b} t \right]^{-2}. \quad (15)$$

For the saved test with $A_b = 200$ m², $h_0 = 0.02$ m, $a = 0.06$ m, and duration 1 h, the exact final depth is 0.015394949 m. Relative numerical error decreases from 0.04004% at 30 s to 0.02004% at 15 s and 0.01003% at 7.5 s. This is consistent with the expected convergence of the chosen first-order time integration.

The companion workbook independently implements the runoff equations, a blocked-storage fixture, and 45 bisection iterations for a controlled-outlet step. The saved artifact audit reports that all 356 workbook formulas were re-evaluated and their cached values checked. These checks are automated; they do not claim a manual Excel session.

6.2 Time-step refinement

The full 216-case grid was repeated at 30, 15, 7.5, 3.75, and 1.875 s. For a metric M , the reported relative change is

$$\delta_M = \frac{|M_{\text{coarse}} - M_{\text{fine}}|}{\max(|M_{\text{fine}}|, 10^{-12})}. \quad (16)$$

Table 9 gives the largest changes across the entire grid for each successive pair. Different maxima may occur in different cases.

Table 9: Grid-wide time-step convergence. Peak and storage changes are percentages; overflow and drawdown changes are absolute.

Steps (s)	Peak change (%)	Storage change (%)	Overflow change (m ³)	Drawdown change (s)
30 / 15	9.0505	0.2712	0.437721	67.253
15 / 7.5	3.5158	0.1367	0.217996	33.573
7.5 / 3.75	3.4323	0.0686	0.109471	16.772
3.75 / 1.875	0.9078	0.0342	0.054714	8.389

Source: Bundled source_data/convergence.csv.

The final maximum downstream-peak change is 0.9078%; maximum-storage change is 0.0342%. Baseline-peak change is at most 0.0472%. All are below 1%. The winner, event pass/fail decisions, and overflow classifications are stable at the final pair. Maximum final overflow-onset movement is 1.875 s; drawdown movement is 8.389 s. Coarser resolutions were insufficient for some overflowing alternatives, even though the design decision was already stable.

These results quantify discretization sensitivity for the stated model. They do not establish an error bound relative to measured site behavior.

6.3 Water-balance closure and reproducibility

The largest recorded absolute residuals across nominal and sensitivity cases are 1.42×10^{-14} m³ for runoff generation, 2.84×10^{-14} m³ for hydrograph transformation, and 1.00×10^{-9} m³ for storage routing. The largest relative routing residual is 5.36×10^{-12} , well below the configured 0.1% relative tolerance with a 10^{-8} m³ absolute allowance.

The existing reproducibility record reports a complete rerun with exact agreement in every compared non-runtime field of the 216-row scenario table, the nine-row baseline table, and the 27-row sensitivity table, as well as an identical selection. During preparation of this report, current project source hashes were checked against that run's manifest and all matched. The new report uses those saved results; it does not present a newly executed full scenario sweep as additional evidence.[8]

7 Sensitivity Results

Table 10 summarizes each one-at-a-time parameter variation across the three design storm shapes. The geometry remains fixed at 150 m³ with the 100 mm opening. Each column reports its own worst case, so the minimum peak reduction, maximum overflow, and maximum drawdown in a row need not occur in the same shape.

Table 10: Selected-design sensitivity for the 50 mm events. Nominal parameter values repeat intentionally in each group.

Parameter	Value	Min. reduction (%)	Max. overflow (m ³)	Max. drawdown (h)	All shapes
Curve number	95	64.37	0.000	7.52	Pass
Curve number	98	63.09	0.000	7.81	Pass
Curve number	100	47.25	2.841	7.95	Fail
Lag (min)	5	62.47	0.000	7.71	Pass
Lag (min)	10	63.09	0.000	7.81	Pass
Lag (min)	15	63.40	0.000	7.90	Pass
C_d	0.5	69.16	0.000	9.72	Pass
C_d	0.62	63.09	0.000	7.81	Pass
C_d	0.75	57.01	0.000	6.42	Pass

Source: Bundled source_data/sensitivity.csv.

Curve number is the tested assumption that changes feasibility. CN 95 and CN 98 pass all three shapes, while CN 100 fails for the back-loaded storm. That case produces 2.841 m³ of overflow and only 47.25% peak reduction. With CN 100, all 50 mm of parking rainfall becomes modeled runoff, so total input is 210 m³, about 22.90 m³ greater than the nominal input. The routed outcome demonstrates that the selected design is not robust to every runoff assumption tested.

Changing lag from 5 to 15 min preserves feasibility throughout this one-at-a-time experiment. Worst-case peak reductions stay between 62.47% and 63.40%, and drawdown remains under 7.90 h. This comparatively small response applies to the tested range and fixed storm duration.

The discharge coefficient shows the expected storage–release tradeoff. Reducing C_d to 0.50 improves worst-case peak attenuation to 69.16% but extends maximum drawdown to 9.72 h. Increasing it to 0.75 shortens drawdown to 6.42 h while reducing worst-case attenuation to 57.01%. Both still meet the nominal objectives, and neither produces overflow in these scenarios.

The experiments establish where the selected design is sensitive within the supplied parameter grid. They do not provide a probability of failure: the parameter values are illustrative, only one parameter changes at a time, and no measured distribution or joint uncertainty model is available.

8 Engineering Discussion and Conclusions

8.1 What the design comparison establishes

The project meets its central analytical goal: it identifies a reproducible minimum among the *tested* feasible capacities and explains the mechanism behind that choice. The 150 m³/100 mm configuration balances enough temporary storage with enough controlled release to avoid overflow across the nominal 50 mm storm shapes. Its smallest peak reduction exceeds the target by 13.09 percentage points, and its longest drawdown is comfortably below 24 h under the adopted assumptions.

The storage value is not a continuously optimized minimum. Candidate capacities are separated by 30 m³, and openings are sampled at 20 mm increments. A capacity between the tested values, or a different outlet geometry, may change the minimum. The current result should therefore be stated as the smallest *tested passing* design, not the unique global optimum or a least-cost design.

The comparison also requires care when interpreting drainage. At the same 100 mm opening, 180 m³ of capacity gives the same nominal 50 mm hydrograph as 150 m³, because neither capacity binds the water level. Its reported drawdown is shorter because 1% of 180 m³ is 1.80 m³, a higher threshold than 1.50 m³. That difference does not demonstrate a faster physical release rate. Capacity-dependent recovery criteria should be kept visible when alternatives are compared.

8.2 Limitations and appropriate next engineering steps

The most important limitation is the gap between a verified numerical implementation and a validated site model. No field survey, soil study, rainfall-runoff observations, or calibrated outlet rating was provided. The synthetic storms are not assigned a frequency, and the acceptance criteria are not a regulatory compliance assessment. The constant-area basin and free-discharge opening omit grading, tailwater, conveyance, freeboard, and overflow-path hydraulics.

The saved results identify two concrete priorities for further work. First, the CN 100 back-loaded failure warrants refining the design under more conservative runoff assumptions before asserting a robust recommendation. Second, the substantial 100 mm overflow shows the need to evaluate an engineered exceedance route if the concept is applied to an actual site. Numerical overflow volumes alone do not locate or size that route.

A subsequent design phase would replace the assumed site geometry and hydrologic inputs with appropriate local information, verify the design storm basis, and assess a realistic stage-storage relationship and outlet system. It could then refine the capacity and opening grid, test combined uncertainty, include tailwater and partial blockage, and examine closely spaced storms with residual initial storage. Cost and maintainability would be additional decision criteria in that phase. These activities are proposed extensions; they are not completed results of this project.

8.3 Final conclusions

1. A 150 m³ basin with a 100 × 100 mm square opening is the smallest tested configuration meeting all three objectives for every nominal 50 mm storm shape.
2. The selected design reduces peak flow by 63.09–74.39%, avoids overflow, and reaches the stated drainage threshold within 7.81 h after rainfall ends. Its maximum simulated design-event depth is 0.6888 m.
3. The selection depends on the stated conditions. All 100 mm storm shapes overflow substantially, and the CN 100 back-loaded 50 mm case fails the overflow and peak criteria.
4. Numerical tests, independent calculations, water balance, full-grid refinement, and recorded reproducibility support the computational findings. They establish numerical reliability within the conceptual model's scope.
5. The modeled benefit is temporary storage and peak attenuation. With zero basin infiltration and evaporation, the incoming runoff remains accounted for as controlled discharge, overflow, or residual storage.

A Complete Nominal Design Comparison

Every row summarizes the three 50 mm storm shapes for one capacity–opening combination. Minimum reduction, maximum overflow, and maximum drawdown are taken separately across shapes. Passing requires all criteria in all three events. Opening dimensions are square side lengths.

Table 11: All 24 tested designs, final time step 1.875 s.

Capacity (m ³)	Depth (m)	Opening (mm)	Min. reduction (%)	Max. overflow (m ³)	Max. drawdown (h)	Result
30	0.15	40	0.00	150.518	37.81	Fail
30	0.15	60	0.00	143.966	24.52	Fail
30	0.15	80	0.00	136.195	18.26	Fail
30	0.15	100	0.00	127.753	14.61	Fail
60	0.30	40	0.00	118.358	30.57	Fail
60	0.30	60	0.00	109.490	18.78	Fail
60	0.30	80	0.00	98.708	13.61	Fail
60	0.30	100	0.00	86.586	10.72	Fail
90	0.45	40	0.00	87.082	28.67	Fail
90	0.45	60	0.00	76.995	16.81	Fail
90	0.45	80	0.00	64.800	11.87	Fail
90	0.45	100	0.00	51.219	9.19	Fail
120	0.60	40	0.00	56.237	28.30	Fail
120	0.60	60	0.00	45.442	15.98	Fail
120	0.60	80	0.00	32.549	11.02	Fail
120	0.60	100	7.49	18.459	8.41	Fail
150	0.75	40	7.87	25.671	28.57	Fail
150	0.75	60	21.46	14.501	15.64	Fail
150	0.75	80	66.92	1.441	10.57	Fail
150	0.75	100	63.09	0.000	7.81	Selected
180	0.90	40	92.96	0.000	28.84	Fail
180	0.90	60	84.87	0.000	15.08	Pass
180	0.90	80	74.63	0.000	9.84	Pass
180	0.90	100	63.09	0.000	7.20	Pass

Source: Bundled source_data/design_selection.csv.

Small values near zero peak reduction are rounded to zero for readability. No candidate below 150 m³ passes. The selected design is the sole passing 150 m³ alternative, so the secondary tie rules do not change the winner.

B Reproducibility, Deliverables, and Contribution Record

B.1 Reproducing the numerical study

The recorded run used Python 3.14.3 on macOS arm64. Key recorded package versions were NumPy 2.4.3, pandas 2.3.3, SciPy 1.17.1, Matplotlib 3.10.9, Plotly 6.7.0, openpyxl 3.1.5, XlsxWriter 3.2.9, and PyMuPDF 1.28.0. The project includes local input files and does not make network requests during a study run. The root `requirements.txt` lists dependency requirements; pdfLaTeX is used for the original report build.

From the project root on the documented machine, the complete original workflow can be run with:

```
conda run -n psrc python run_project.py
```

The default command regenerates the original project outputs. To preserve the saved study while running a changed configuration, use a separate output directory, for example:

```
conda run -n psrc python run_project.py \
  --config inputs/my_experiment.json \
  --output-dir experiments/my_experiment
```

The alternative configuration must first be created by copying and editing the supplied input file. This command illustrates reproduction and extension; it does not imply that such an additional experiment was performed for this report. Model tests can be run independently with `python -m unittest discover -s tests -v`.

The recorded study stage took 301.60 s, and the complete recorded command took 314.25 s. Median final-resolution case time was 0.3663 s, and the slowest case took 1.1556 s. Recorded peak Python-process memory was 408.52 MiB, excluding child processes. These are measurements from the saved run on that machine, rather than general performance guarantees or timing for this expanded report build.

B.2 Project deliverables and this report package

Table 12: Deliverables produced by the project and their intended uses.

File or directory at the project root	Purpose
<code>stormwater_dashboard.html</code>	Offline exploration of all 216 cases and the matching baselines.
<code>stormwater_calculations.xlsx</code>	Scenario tables, formula-based independent checks, and charts.
<code>site_plan.pdf</code> , <code>site_plan.svg</code> , <code>site_plan.dxf</code>	Conceptual plan and section, including editable drawing formats.
<code>results/</code>	Saved metrics, full time-series arrays, source snapshots, manifest, convergence, and verification records.
<code>engineering_report.tex</code> and PDF	Original brief generated report.
<code>FINDINGS.md</code> , <code>LEARNING_GUIDE.md</code>	Findings and worked learning material.
<code>manasi_project_report/</code>	This expanded engineering report, editable LaTeX, PDF, figure assets, numerical tables, and evidence snapshots.

This report folder is self-contained for LaTeX compilation. Run `sh build_report.sh` inside it to rebuild the PDF. The included `generate_tables.py` uses only Python’s standard library and rebuilds tables

from the bundled evidence snapshot. Rebuilding tables does not rerun the engineering model or revise the narrative automatically. If new model results are adopted later, the source snapshot, tables, figures, and written interpretation should be updated together.

B.3 Evidence and actual contributions

The supplied project plan defined the technical scope. Codex implemented and executed the numerical study, developed the automated checks, and created the workbook, dashboard, figures, drawings, and documentation. This expanded report was prepared from the saved project evidence with Codex assistance.

The project contribution record does not verify Manasi's personal coding, manual Excel calculations, CAD edits, or independent interpretation. Those activities remain for her to record in `MANASI_REVIEW.md`; the report does not sign or complete that review on her behalf. This distinction preserves an accurate record of the completed technical work and the remaining individual learning activities.^[9]

The new package's `source_data/report_provenance.json` identifies its copied evidence and source-hash checks. The original artifact audit and reproducibility records refer to the original project deliverables. Compilation and layout checks for this expanded report are recorded separately under `build/`.

References

- [1] *Parking-Lot Stormwater Detention: Simulation and Design Comparison*. Local project specification, PROJECT_PLAN.md, completed-study status dated September 8, 2026. Read together with README.md and FINDINGS.md in the project root.
- [2] U.S. Army Corps of Engineers, Hydrologic Engineering Center. *SCS Curve Number Loss Model*, HEC-HMS Technical Reference Manual. Accessed September 8, 2026. Basis for cumulative event runoff equations and incremental excess rainfall.
- [3] U.S. Army Corps of Engineers, Hydrologic Engineering Center. *SCS Unit Hydrograph Model*, HEC-HMS Technical Reference Manual. Accessed September 8, 2026. Basis for dimensionless hydrograph interpretation and $T_p = t_{lag} + \Delta t/2$.
- [4] Texas Department of Transportation. *NRCS Dimensionless Unit Hydrograph*, Hydraulic Design Manual, Chapter 4, Section 13, Table 4-29. Accessed September 8, 2026. Source of the 33 reference ordinate pairs stored with the project.
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- [6] U.S. Army Corps of Engineers, Hydrologic Engineering Center. *Outflow Structures Routing*, HEC-HMS User's Manual. Accessed September 8, 2026. Background on outlet regimes and limitations; the integrated square-opening rating used here is a project idealization.
- [7] Texas Department of Transportation. *Selection of the Appropriate Method for Calculating Runoff*, Hydraulic Design Manual, Chapter 4, Section 7. Accessed September 8, 2026. Background for using a hydrograph when evaluating storage.
- [8] *Manasi project numerical results and verification evidence*. Saved run completed September 8, 2026, 21:44:16 UTC. results/scenarios.csv, results/baselines.csv, results/design_selection.csv, results/selection.json, results/convergence.csv, results/sensitivity.csv, results/independent_checks.json, results/run_manifest.json, results/artifact_audit.json, and results/reproducibility_check.json. Copies supporting this report are bundled in source_data/.
- [9] *Contribution and Assistance Record* and *Manasi's Personal Review*. Local project files CONTRIBUTIONS.md and MANASI_REVIEW.md, reviewed for this report on September 8, 2026.